First-step analysis is the single most reusable trick in Stage-1 stochastic questions: *look only at the first move, write what happens next in terms of the same unknowns, and solve a tiny linear system.* For expected times the system is $k_i = 1 + \sum_j P_{ij} k_j$ (the "1+" pays for the step you take), for hitting probabilities it is $h_i = \sum_j P_{ij} h_j$ with boundary values 0 and 1. Every problem is color-coded and carries an analogy; Section C then packages everything into the fundamental matrix $N = (I - Q)^{-1}$.

A. Expected Hitting & Absorption Times

Problem A1 — The simplest first-step equation

A day is Sunny with $P(\text{Sunny} \to \text{Rainy}) = 0.2$. Starting sunny, find the expected number of days until the first rainy day.

Solution. Let k be the expected wait. After one day: with probability 0.2 we are done, with probability 0.8 we are back where we started:

$$k = 1 + 0.8\,k + 0.2 \cdot 0 \;\Rightarrow\; 0.2k = 1 \;\Rightarrow\; k = \boxed{5}.$$

★ *Intuition:* This is a geometric waiting time written as a one-line equation. The "1+" is the toll for the step you take; the remaining terms are "what is left to do after paying it." Every longer first-step equation is this idea with more rooms.

Problem A2 — Three states, one target

$P_{1\cdot} = (0.5, 0.5, 0)$, $P_{2\cdot} = (0.3, 0.4, 0.3)$, and state 3 is the target. Find the expected time to reach state 3 from states 1 and 2.

Solution. With $k_3 = 0$:

$$k_1 = 1 + 0.5k_1 + 0.5k_2, \qquad k_2 = 1 + 0.3k_1 + 0.4k_2.$$

The first gives $k_1 = 2 + k_2$. Substituting: $0.6k_2 = 1 + 0.3(2 + k_2) \Rightarrow 0.3k_2 = 1.6$:

$$k_2 = \boxed{\tfrac{16}{3} \approx 5.33}, \qquad k_1 = \boxed{\tfrac{22}{3} \approx 7.33}.$$

★ *Intuition:* State 1 can only reach the target *through* state 2, so $k_1 = k_2 + 2$: the extra 2 is the expected number of attempts needed to move from 1 to 2 (success probability 0.5 per step). Reading the diagram before writing equations often gives you such relationships for free.

▶ *Speed trick:* Eliminate the state farthest from the target first — here k_1 is a one-line expression in k_2.

Problem A3 — Waiting for the pattern HTH

A fair coin is flipped repeatedly. Find the expected number of flips until the pattern HTH first appears.

Solution. Track the longest useful suffix: $\varnothing$, H, HT. With e_s the expected remaining flips:

$$e_\varnothing = 1 + \tfrac{1}{2}e_H + \tfrac{1}{2}e_\varnothing, \qquad e_H = 1 + \tfrac{1}{2}e_H + \tfrac{1}{2}e_{HT},$$

$$e_{HT} = 1 + \tfrac{1}{2}(0) + \tfrac{1}{2}e_\varnothing.$$

Solving, $e_{HT} = 6$, $e_H = 8$, and $e_\varnothing = \boxed{10}$.

★ *Intuition:* After HT, the next flip either finishes the pattern (H) or sends you all the way back (T gives "HTT," with no useful suffix). Patterns that *overlap with themselves* are harder to finish: HTH needs 10, while HHT needs only 8, HH needs 6, HT needs 4, and HHH needs 14.

▶ *Speed trick:* For a pattern of length ℓ, the expected waiting time is $\sum_k 2^k$ over every k for which the prefix of length k equals the suffix of length k. HTH: $2^3 + 2^1 = 10$.

Problem A4 — Duration of a fair gambler's ruin

A fair walk starts at 3 on $\{0, 1, \ldots, 10\}$ with absorbing ends. Find the expected number of steps to absorption.

Solution. The system is $k_i = 1 + \frac{1}{2}k_{i-1} + \frac{1}{2}k_{i+1}$, i.e. $k_{i+1} - 2k_i + k_{i-1} = -2$ with $k_0 = k_{10} = 0$. A constant second difference means a quadratic, and the boundary conditions give

$$k_i = i(N - i) \;\Rightarrow\; k_3 = 3(10 - 3) = \boxed{21}.$$

★ *Intuition:* A drunkard on a plank takes about as many steps as the *product* of the distances to the two edges — not their sum. Doubling the plank length quadruples the time, because the walker needs d^2 steps to travel distance d in a random walk.

▶ *Speed trick:* Fair ruin: duration $= k(N - k)$, hitting probability $= k/N$. Memorize both together.

Problem A5 — Duration of a biased gambler's ruin

A gambler wins each \$1 bet with probability $p = 0.6$ ($q = 0.4$), starts with \$2 and stops at \$0 or \$5. Find the expected number of bets.

Solution. With $r = q/p = \frac{2}{3}$, the standard duration formula is

$$D_k = \frac{k}{q - p} - \frac{N}{q - p} \cdot \frac{1 - r^k}{1 - r^N}.$$

With $q - p = -0.2$, $k = 2$, $N = 5$:

$$D_2 = -10 + 25 \cdot \frac{1 - \frac{4}{9}}{1 - \frac{32}{243}} \approx -10 + 25(0.6398) \approx \boxed{5.995}.$$

(Checked against solving the full 4×4 system, which gives 5.9953.)

★ *Intuition:* The fair walk from \$2 of \$5 would take $2 \cdot 3 = 6$ bets; the 60% edge shortens that only slightly (5.995) because the walls are so close that the drift has little room to act. The gap between the fair and biased durations grows quickly as the walls move apart.

Problem A6 — Hitting the opposite vertex of a hexagon

A walker moves to a uniformly random neighbour on a 6-cycle. Starting at vertex 0, find the expected time to reach vertex 3 (the opposite vertex).

Solution. The two arcs from 0 to 3 are mirror images, so only the walker's *distance d from the start* matters, with absorption when $d = 3$. The moves are: $d = 0 \to 1$ with probability 1 (both neighbours of 0 are at distance 1); $d = 1 \to 0$ or 2 w.p. $\frac{1}{2}$ each; $d = 2 \to 1$ or 3 w.p. $\frac{1}{2}$ each. With k_d the expected remaining time:

$$k_0 = 1 + k_1, \qquad k_1 = 1 + \tfrac{1}{2}k_0 + \tfrac{1}{2}k_2,$$

$$k_2 = 1 + \tfrac{1}{2}k_1 + \tfrac{1}{2}(0).$$

Substituting $k_2 = 1 + \frac{1}{2}k_1$ and $k_0 = 1 + k_1$: $k_1 = 2 + \frac{3}{4}k_1 \Rightarrow k_1 = 8$, so

$$k_0 = \boxed{9}.$$

★ *Intuition:* On an n-cycle the expected time to hit the antipodal vertex is $(n/2)^2 = 9$ — the same as a fair gambler's ruin with walls 3 steps away on each side. Symmetry folds the cycle into a line, and the line obeys the $k(N-k)$ rule: the start is the reflecting end and the target is the wall, which gives $3^2 = 9$.

B. Probability of Reaching A Before B

Problem B1 — A casino's biased edge

A gambler starts with \$10, bets \$1 at a time with win probability $p = 0.45$ and plays until \$20 or \$0. Find $P(\text{reach \$20})$.

Solution. With $r = q/p = \frac{0.55}{0.45} = \frac{11}{9}$, boundary conditions $h_0 = 0$, $h_{20} = 1$:

$$h_{10} = \frac{1 - r^{10}}{1 - r^{20}} = \frac{1}{1 + r^{10}} = \frac{1}{1 + 7.45} \approx \boxed{0.1185}.$$

★ *Intuition:* A 5% disadvantage per bet becomes an 88% probability of failing to double up: you must out-run a persistent drift for ten net wins. A small edge is a tilt, and over long runs a tilted table sends everything downhill.

▶ *Speed trick:* Starting in the *middle* of a symmetric interval $[0, 2k]$ with ratio r: $h = \frac{1}{1+r^k}$.

Problem B2 — A graph version

A walker moves to a uniformly random neighbour on the graph with edges $1{-}2$, $2{-}3$, $3{-}4$, $4{-}1$, $1{-}3$. Starting at 2, find the probability of reaching 4 before 1.

Solution. Targets: $h_4 = 1$, $h_1 = 0$. From 2 the neighbours are $1, 3$; from 3 they are $1, 2, 4$:

$$h_2 = \tfrac{1}{2}h_1 + \tfrac{1}{2}h_3 = \tfrac{1}{2}h_3,$$

$$h_3 = \tfrac{1}{3}h_1 + \tfrac{1}{3}h_2 + \tfrac{1}{3}h_4 = \tfrac{1}{3}h_2 + \tfrac{1}{3}.$$

So $h_3 = \frac{1}{6}h_3 + \frac{1}{3} \Rightarrow h_3 = \frac{2}{5}$ and $h_2 = \boxed{\frac{1}{5}}$.

★ *Intuition:* From node 2 you must first reach node 3 to even have a chance at 4, and node 1 (the "bad" neighbour) is adjacent to both. Writing equations only for the *unknown* nodes keeps the system as small as possible.

Problem B3 — Penney's game: which pattern wins?

Flip a fair coin until either THH or HHT appears. Find $P(\text{THH appears first})$.

Solution. States by useful suffix: $\varnothing$, H, T, HH, TH, with $h_s = P(\text{THH first})$:

$$h_{HH} = \tfrac{1}{2}h_{HH} + \tfrac{1}{2}(0) \Rightarrow h_{HH} = 0,$$

$$h_{TH} = \tfrac{1}{2}(1) + \tfrac{1}{2}h_T, \qquad h_T = \tfrac{1}{2}h_T + \tfrac{1}{2}h_{TH} \Rightarrow h_T = h_{TH} = 1.$$

$$h_H = \tfrac{1}{2}h_{HH} + \tfrac{1}{2}h_T = \tfrac{1}{2}, \quad h_\varnothing = \tfrac{1}{2}h_H + \tfrac{1}{2}h_T = \boxed{\tfrac{3}{4}}.$$

★ *Intuition:* Once HH has been seen, the only way out is T, which *completes HHT* — so after HH, THH can never win. But once T has been seen, THH is guaranteed to beat HHT: the T keeps resetting, and eventually an H follows. Pattern games are non-transitive precisely because patterns can "poison" one another's progress.

Problem B4 — Holding probabilities don't change hitting probabilities

On $\{0, 1, 2, 3\}$ (absorbing $0, 3$): from 1 stay w.p. 0.4, up 0.3, down 0.3; from 2 stay w.p. 0.2, up 0.4, down 0.4. Find $P(\text{hit 3 before 0} \mid \text{start 1})$.

Solution. $h_1 = 0.4h_1 + 0.3h_2$ and $h_2 = 0.2h_2 + 0.4 + 0.4h_1$. So $h_2 = 0.5 + 0.5h_1$ and

$$0.6h_1 = 0.3(0.5 + 0.5h_1) \Rightarrow 0.45h_1 = 0.15 \Rightarrow h_1 = \boxed{\tfrac{1}{3}}.$$

★ *Intuition:* Staying put wastes *time* but not *direction*. Conditional on moving, each state is a fair coin, so the answer is the gambler's-ruin value $\frac{1}{3}$ — the same as the lazy-free walk. Holding times appear in expected times (they stretch them by $\frac{1}{1-\text{stay}}$) but vanish from hitting probabilities.

▶ *Speed trick:* Ignore self-loops when computing *probabilities*; divide the other probabilities by $1 - P_{ii}$. Include them when computing *times*.

Problem B5 — Winning a tennis game from deuce

The server wins each point w.p. $p = 0.6$. From deuce, a player must win by two points. Find $P(\text{server wins the game})$.

Solution. States: Deuce D, server-advantage A_S, receiver-advantage A_R:

$$h_D = p\,h_{A_S} + q\,h_{A_R}, \quad h_{A_S} = p + q\,h_D, \quad h_{A_R} = p\,h_D.$$

So $h_D = p^2 + 2pq\,h_D$:

$$h_D = \frac{p^2}{1 - 2pq} = \frac{p^2}{p^2 + q^2} = \frac{0.36}{0.52} \approx \boxed{0.692}.$$

★ *Intuition:* A 60% point-winner wins nearly 70% of deuce games: "win by two" is a magnifier. Every extra point you must win in a row amplifies the better player's edge, which is why tennis, with its multi-level scoring, is so reliably won by the better player.

C. Absorbing Chains & the Fundamental Matrix

Problem C1 — Canonical form and $B = NR$

States $1, 4$ are absorbing; transient states $2, 3$ have $P_{2\cdot} = (0.3, 0.4, 0.3, 0)$, $P_{3\cdot} = (0, 0.2, 0.5, 0.3)$. Find $N = (I - Q)^{-1}$ and the absorption probabilities $B = NR$.

Solution. Order transient states first: $Q = \begin{pmatrix} 0.4 & 0.3 \\ 0.2 & 0.5 \end{pmatrix}$, $R = \begin{pmatrix} 0.3 & 0 \\ 0 & 0.3 \end{pmatrix}$ (columns: absorb at 1, absorb at 4).

$$N = \frac{1}{0.24}\begin{pmatrix} 0.5 & 0.3 \\ 0.2 & 0.6 \end{pmatrix} = \begin{pmatrix} 2.083 & 1.25 \\ 0.833 & 2.5 \end{pmatrix}.$$

$$B = NR = \begin{pmatrix} 0.625 & 0.375 \\ 0.25 & 0.75 \end{pmatrix}.$$

★ *Intuition:* N_{ij} is the expected number of visits to room j before escaping, starting in room i. Multiplying by R converts "visits" into "escapes": each visit to room j gives a chance R_{jk} of leaving through exit k.

▶ *Speed trick:* $B = NR$ answers every "which absorbing state" question at once; row sums of N give expected absorption times.

Problem C2 — The variance of the absorption time

For the fair gambler on $\{0, \ldots, 4\}$ starting at 2, find $\mathbb{E}[T]$ and $\mathrm{Var}(T)$ using $\mathrm{Var}(T) = (2N - I)t - t \circ t$, where $t = N\mathbf{1}$.

Solution. With Q over states $1, 2, 3$, $N = \begin{pmatrix} 1.5 & 1 & 0.5 \\ 1 & 2 & 1 \\ 0.5 & 1 & 1.5 \end{pmatrix}$ and $t = (3, 4, 3)$. For the middle state:

$$[(2N - I)t]_2 = (2, 3, 2) \cdot (3, 4, 3) = 24, \qquad t_2^2 = 16,$$

$$\mathbb{E}[T] = \boxed{4}, \qquad \mathrm{Var}(T) = 24 - 16 = \boxed{8} \quad (\text{sd} \approx 2.83).$$

★ *Intuition:* The expected time is 4 steps, but the typical wobble is nearly 3 steps either way: a walk can end in 2 steps (two wins in a row) or drag on for 10. Knowing the variance warns you that "about 4" is far from certain.

Problem C3 — A maze with three exits

Two rooms: Room 1 stays w.p. 0.2, goes to Room 2 w.p. 0.3, exits to A w.p. 0.5. Room 2 goes to Room 1 w.p. 0.25, stays w.p. 0.25, exits to B w.p. 0.3 and to C w.p. 0.2. From Room 1, find the exit probabilities and the expected escape time.

Solution. $Q = \begin{pmatrix} 0.2 & 0.3 \\ 0.25 & 0.25 \end{pmatrix}$, $R = \begin{pmatrix} 0.5 & 0 & 0 \\ 0 & 0.3 & 0.2 \end{pmatrix}$, and

$$N = (I-Q)^{-1} = \begin{pmatrix} \frac{10}{7} & \frac{4}{7} \\ \frac{10}{21} & \frac{32}{21} \end{pmatrix}.$$

First row of $B = NR$:

$$(P_A, P_B, P_C) = \left(\tfrac{5}{7},\ \tfrac{6}{35},\ \tfrac{4}{35}\right) \approx (0.714,\ 0.171,\ 0.114).$$

$$\mathbb{E}[T] = \tfrac{10}{7} + \tfrac{4}{7} = \boxed{2}.$$

★ *Intuition:* Room 1 has a direct exit A; the only way to reach B or C is to pass through Room 2 first. The numbers reflect it: Room 1's escape is dominated by exit A (71%), with the other two exits sharing the remaining 29%.

Problem C4 — Return probability and "ever visit" probability

Using C1's chain, find the probability of ever returning to state 2 from state 2, and the probability of ever visiting state 3 from state 2.

Solution. For transient states,

$$f_{jj} = 1 - \frac{1}{N_{jj}} = 1 - \frac{1}{2.083} = \boxed{0.52},$$

$$P(i \text{ ever visits } j) = \frac{N_{ij}}{N_{jj}} = \frac{1.25}{2.5} = \boxed{0.5}.$$

★ *Intuition:* Visits to a room come in a geometric burst: each visit is followed by another with probability f_{jj}, so the mean number is $\frac{1}{1-f_{jj}} = N_{jj}$. Dividing N_{ij} by N_{jj} strips away the burst and leaves only the chance that the burst starts at all.

D. Setting Up Systems Under Time Pressure

Problem D1 — Two sixes in a row

A fair die is rolled until two consecutive sixes appear. Find the expected number of rolls.

Solution. States: 0 (last roll not a six) and 1 (last roll a six):

$$e_0 = 1 + \tfrac{1}{6}e_1 + \tfrac{5}{6}e_0, \qquad e_1 = 1 + \tfrac{1}{6}(0) + \tfrac{5}{6}e_0.$$

Substituting: $e_0 = 6 + e_1$ and $e_1 = 1 + \frac{5}{6}e_0 \Rightarrow e_0 = \boxed{42}$.

★ *Intuition:* It takes 6 rolls to see *one* six, but a second consecutive six needs $6^2 + 6 = 42$: each failure sends you back to the start. Waiting for r consecutive successes with probability p takes $\frac{p^{-r}-1}{1-p}$ trials.

▶ *Speed trick:* Waiting for r successes in a row: $\mathbb{E} = \dfrac{(1/p)^r - 1}{1-p}$.

Check: $p = \frac{1}{2}, r = 2 \Rightarrow \frac{4-1}{0.5} = 6$; $p = \frac{1}{6}, r = 2 \Rightarrow \frac{36-1}{5/6} = 42$.

Problem D2 — Hitting a pendant vertex

A walker moves to a uniformly random neighbour on the graph with a triangle $1, 2, 3$ and a pendant vertex 4 attached to 3. Find the expected time to reach vertex 4 from vertex 1.

Solution. By symmetry $k_1 = k_2 = a$. Degrees: vertices $1, 2$ have 2 neighbours; vertex 3 has 3.

$$a = 1 + \tfrac{1}{2}a + \tfrac{1}{2}k_3, \qquad k_3 = 1 + \tfrac{1}{3}(2a) + \tfrac{1}{3}(0).$$

The first gives $a = 2 + k_3$. Substituting: $k_3 = 1 + \frac{2}{3}(2 + k_3) \Rightarrow$ $k_3 = 7$, so

$$k_1 = \boxed{9} \text{ (and } k_3 = 7).$$

★ *Intuition:* The triangle is a "trap" that keeps the walker bouncing around before it finds the only exit at 3 and then a $\frac{1}{3}$ chance to leave toward 4. More edges around the exit door means a longer wait.

Problem D3 — Ruin with ties

A game round is won w.p. 0.4, lost w.p. 0.4 and tied w.p. 0.2 (no change). Starting at \$2 with absorbing barriers \$0 and \$4, find the expected number of rounds.

Solution. Ignoring ties, the fair walk takes $k(N-k) = 4$ *moves*. Each move requires on average $\frac{1}{0.8}$ rounds (ties are geometric delays):

$$\mathbb{E}[\text{rounds}] = \frac{4}{0.8} = \boxed{5}.$$

★ *Intuition:* Ties are traffic lights: they cost time but never change the route. The route (and thus the win probability, still $\frac{1}{2}$) is the same as the tie-free game; the time is stretched by $\frac{1}{1-0.2}$.

E. Quant Capstones

Problem E1 — A trader with a stop-loss and a target

A trader nets +\$1k per trade w.p. 0.55 and −\$1k w.p. 0.45. She stops at a loss of \$2k or a profit of \$3k. Find $P(\text{hit the profit target})$ and the expected number of trades.

Solution. Shift to $\{0, \dots, 5\}$ with start 2, $r = q/p = \frac{9}{11}$:

$$P(\text{target}) = \frac{1-r^2}{1-r^5} = \frac{0.3306}{0.6334} \approx \boxed{0.522}.$$

$$\mathbb{E}[\text{trades}] = \frac{k}{q-p} - \frac{N}{q-p} \cdot P = -20 + 50(0.522) \approx \boxed{6.10}.$$

★ *Intuition:* A modest 55% win rate and a 3:2 reward-to-risk ratio gives only a 52% chance of hitting the target. The target is farther away than the stop, which offsets the edge. A trader can only claim a "real" edge after accounting for how the *boundaries* are placed.

▶ *Speed trick:* Fair-odds sanity check: with $p = \frac{1}{2}$ the target probability would be $\frac{2}{5} = 0.40$. The 55% edge lifts it by 12 points, not by 5.

Problem E2 — How long does a deuce last?

Using B5's setup ($p = 0.6$), find the expected number of additional points played from deuce.

Solution. From deuce, two consecutive points either end the game (probability $p^2 + q^2 = 0.52$) or return to deuce. The number of two-point blocks is geometric with mean $\frac{1}{0.52}$:

$$\mathbb{E}[\text{points}] = \frac{2}{p^2+q^2} = \frac{2}{0.52} \approx \boxed{3.85}.$$

★ *Intuition:* Deuce is a coin flip repeated in pairs: each pair either "settles" the game or resets it. The more evenly matched the players, the more often a pair splits 1–1 and the longer a deuce lasts; at $p = \frac{1}{2}$ the average is exactly 4 points.

▶ *Speed trick:* $p^2 + q^2 = 1 - 2pq$: the settling probability per pair is $1 - 2pq$, maximal when p is far from $\frac{1}{2}$.

Drill order for review: re-derive A3 (pattern waiting time), A4 ($k(N-k)$ from the second-difference equation), B3 (Penney's game) and C1 (N and $B = NR$) from scratch, closed-book. They carry the chapter's central workflow: pick the state space (suffixes, distances, positions), write one first-step equation per state ("1+" for times, boundary values for probabilities),

and eliminate from the target backward.